How could the UK regulate cryptoassets to balance consumer protection with fostering a free and accessible market?

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Declaration:

I declare that the work in this Dissertation in Law was carried out in accordance with the Regulations of The University of Law. The work is original except where indicated by acknowledgement or special reference in the text, and no part of this Dissertation in Law has been presented to any other University or body for examination either in the United Kingdom or overseas.
I also declare that I have not conducted research involving human participants.

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1. Introduction

On the 9th of January 2009, the pseudonymous web user Satoshi Nakamoto released the Bitcoin network.[1] By the beginning of 2013, one Bitcoin was worth $13.30.[2] By the end, that figure was $754.22. There was another boom in 2017, when the value shot up from $998.33 to $13,657.20. Today, the market valuation of crypto is over $1 trillion, with Bitcoin making up almost half of that.[3] With this new technology exploding into global relevance, governments the world over have been scrambling to figure out how to manage it.[4] Some countries have taken a strict regulatory approach (China banned all crypto transactions in September 2021[5]), while others have embraced freedom (in that same month El Salvador became the first country to declare Bitcoin legal tender[6]). Many, like the UK, are yet to commit – much remains uncertain about how the crypto markets operate, and the UK government seems to be waiting for the dust to settle before committing. They have released multiple consultation papers, allowing the public and industry to comment. The most recent of these, containing proposals for the regulation of cryptoassets, closed for submissions on April 30th 2023.[7]

Britain has taken a relatively liberal approach to the regulation of financial markets since the "Big Bang" of 1986.[8] In a 2021 speech regarding exit from the EU, Edwin Latter, Director of Markets and Wholesale Policy at the Financial Conduct Authority (FCA), said, "Where regulation has imposed costs without beneficial outcomes to justify that cost, then

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[1] Satoshi Nakamoto, 'Bitcoin v0.1 released' (*Internet Archive*, 9 January 2009) <https://web.archive.org/web/20140326174921/http:/www.mail-archive.com/cryptography%40metzdowd.com/msg10142.html> accessed 8 May 2023.
[2] Ian Webster, 'Bitcoin Historical Prices' (*Official Data*) <www.officialdata.org/bitcoin-price> accessed 8 May 2023.
[3] CoinMarketCap, 'Today's Cryptocurrency Prices by Market Cap' (*CoinMarketCap*) <https://coinmarketcap.com/> accessed 8 May 2023.
[4] —, 'What is the current state of cryptocurrency regulation?' (*World Economic Forum*, 28 Mar 2022) <www.weforum.org/agenda/2022/03/where-is-cryptocurrency-regulation-heading/> accessed 8 May 2023.
[5] —, 'China Declares all crypto-currency transactions illegal' *BBC News* (24 September 2021) <www.bbc.co.uk/news/technology-58678907> accessed 8 May 2023.
[6] Joe Hernandez, 'El Salvador Just Became The First Country To Accept Bitcoin As Legal Tender' *NPR* (7 September 2021) <www.npr.org/2021/09/07/1034838909/bitcoin-el-salvador-legal-tender-official-currency-cryptocurrency> accessed 8 May 2023.
[7] HM Treasury, *Future financial services regulatory regime for cryptoassets* (2023).
[8] Philip Booth, *Regulation Without the State* (IEA Discussion Paper No. 97, 2019) <https://iea.org.uk/wp-content/uploads/2019/07/DP97_Regulation-without-the-state_web.pdf> accessed 8 May 2023.

we will want to use our new ability to change direction."[9] This approach has led to the UK's being one of the global hubs of the finance industry, with London the second most competitive financial centre in the world behind New York.[10]

The same has the potential to be true of the crypto market. Writing in September 2018, the House of Commons Treasury Committee stated:

> "If the UK develops an appropriate and proportionate regulatory environment for crypto-assets and if future innovations in crypto-assets proved themselves as beneficial to society and industry, the UK could be well placed to become a global centre for this activity, providing that the crypto-asset market adhered to high standards and was not associated with criminality."[11]

This essay is an attempt to present a viable and legally grounded framework to fulfil that statement. But what constitutes a "proportionate" regulatory environment? If the benefit of less regulation is the prosperity of the markets, then the benefit of greater regulation is the protection of the public and society at large. As the World Bank describes: "A critical challenge for policy makers is to harness the benefits and opportunities of fintech while managing its risks, including for consumers."[12]

For the purposes of this essay, an 'accessible' cryptoasset market is one in which the technology is readily accessible to the public. This does not just mean the ability to trade in them as investments, but also to use them privately as a medium of exchange. As will be discussed, there are many reasons why a person may wish to use crypto, but also many risks associated with the technology. This essay will discuss whether this is desirable, and explore what direction the UK could take to protect consumers from the dangers of using cryptoassets while still realising the vision of a 'free and accessible' market.

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[9] Edward Schooling Latter, 'A forward look at regulation of the UK's wholesale financial markets' (Speech at the ISLA's 11th Post Trade Conference, Virtual, 16 March 2021) <www.fca.org.uk/news/speeches/forward-look-regulation-uks-wholesale-financial-markets> accessed 8 May 2023.

[10] Mark Yeandle, 'The Global Financial Centres Index 20' (Long Finance & Financial Centre Futures, 2016) <www.longfinance.net/publications/long-finance-reports/the-global-financial-centres-index-20/> accessed 8 May 2023.

[11] House of Commons Treasury Committee, *Crypto-assets* (HC 910, 2018) [105].

[12] World Bank, 'Consumer Risks in Fintech: New Manifestations of Consumer Risks and Emerging Regulatory Approaches' (2021) <https://responsiblefinance.worldbank.org/en/responsible-finance/financial-consumer-protection> accessed 8 May 2023.

2. What is a Cryptoasset?

The first step is to understand what exactly a cryptoasset is. The legal definition in use in the UK is "a cryptographically secured digital representation of value or contractual rights that uses a form of distributed ledger technology (DLT) and can be transferred, stored or traded electronically".[13] Rather than being a digital file, as many might imagine, a cryptoasset is an assignment of value to a 'wallet' on a publicly available ledger[14]. Each wallet consists of a public key and a private key. The public key can be thought of as a username, or an address, against which the ledger records cryptoassets owned. The private key can be thought of as a password, used to verify transactions. The amount allocated to a public key is publicly available knowledge, but only someone with the corresponding private key can transfer those assets to other addresses.[15] Anyone can set up a crypto wallet on their own, without having to register their identity with a bank or database.[16] So while all transactions, and the values of all wallets, are public, the people who own those wallets may not be.

The heart of a cryptoasset system is the cryptographic process that secures it. The mechanics of this go beyond the scope of this essay, but the effect is that it can be independently verified that the person sending the assets has the correct private key without the verifier knowing what the private key actually is.[17] This allows transactions to be monitored and verified in a decentralised manner, without a banking system or government fiat.[18] This was the original intention that drove the development of Bitcoin – to have a medium of trade independent from the established financial system.[19]

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[13] Money Laundering, Terrorist Financing and Transfer of Funds (Information on the Payer) Regulations 2017, SI 2017/692.
[14] Luke Tredinnick, 'Cryptocurrencies and the blockchain' (2019), Business Information Review 36(1), 39–4.
[15] Kirsty Moreland, 'What Are Public Keys Vs Private Keys?' (*Ledger Academy*, 23 October 2019) <www.ledger.com/academy/blockchain/what-are-public-keys-and-private-keys> accessed 8 May 2023.
[16] Dalia Ramirez, 'How to Get a Crypto Wallet' (*nerdwallet*, 21 April 2023) <www.nerdwallet.com/article/investing/get-a-crypto-wallet> accessed 8 May 2023.
[17] Tredinnik (n 14).
[18] Ibid.
[19] Rebecca Baldridge and Bryan Hood, 'As Cryptocurrency Plummets, the Father of Bitcoin Is Nowhere to Be Found' *Robb Report* (2 December 2022) <https://robbreport.com/lifestyle/finance/bitcoin-founder-satoshi-nakamoto-1234613022/> accessed 8 May 2023.

However, this freedom is not without its costs. Because there is no centralised authority, payments are irreversible.[20] Thus, if money is sent to the wrong address, stolen, or appropriated in another manner (such as fraud), the original owner is left with little recourse – though not none.[21]

These three qualities – anonymity, lack of centralisation, and irreversibility – are the most pertinent when one is concerned with the regulation of cryptoassets. They mean that crypto is difficult to control – by design.

3. Ownership of Cryptoassets

The next step is to get a picture of how cryptoasset technology is currently handled in English law. A fundamental question that had to be answered was: are cryptoassets property? Can and should something that only exists by mutual agreement be considered property in law? This was answered in the landmark case *AA v Persons Unknown* (2019).[22] In this case, the claimant had been blackmailed into paying 109.5 Bitcoin to the unknown defendants. A specialist had been able to trace the Bitcoin to an account held at the crypto exchange Bitfinex. The High Court judge granted an interim proprietary injunction on the Bitcoin, preventing it from being transferred away[23] – and in the process, confirmed that "cryptocurrency such as Bitcoin" were a form of property (in order to be subject to the injunction). Brian J said, "They meet the four criteria set out in Lord Wilberforce's classic definition of property in *National Provincial Bank v Ainsworth*[1965] AC 1175 as being definable, identifiable by third parties, capable in their nature of assumption by third parties, and having some degree of permanence."

This seems sensible, courtesy of the public ledger system. The ledger defines a cryptoasset as the base unit of the values it assigns to each address, and the assets can be easily identified by checking those values. A cryptoasset may be transferred to another wallet, and will exist as long as its public ledger does. It may be counterintuitive to accept that information without any physical manifestation (like a bank deposit has with cash), but

[20] Blockchain.com, 'Can my transaction be cancelled or reversed?' (*Blockchain.com*, 7 March 2022) <https://support.blockchain.com/hc/en-us/articles/211162263-Can-my-transaction-be-canceled-or-reversed-> accessed 8 May 2023.
[21] Kevin Groves, 'Can You Recover Lost or Stolen Bitcoin?' (*HedgewithCrypto*, 11 April 2023) <www.hedgewithcrypto.com/recover-lost-or-stolen-bitcoin/> accessed 8 May 2023.
[22] *AA v Persons Unknown & Ors, Re Bitcoin* [2019] EWHC 3556 (Comm), [2020] 4 W.L.R. 35.
[23] This was enforceable as the crypto was held at an exchange, which could cooperate with the courts.

as the New Zealand high court said in *Ruscoe v Cryptopia* (2020),[24] crypto is more than just "pure information", as "the whole purpose behind cryptocurrencies is to create an item of tradeable value, not simply to record or to impart in confidence knowledge or information". Thus cryptoassets have the legal qualities of property in English law. They can be traced and enforced against, as confirmed in *Ion Science v Persons Unknown* (2022),[25] and can be the subject of a trust as confirmed in *Wang v Derby* (2022).[26]

One issue that arises with cryptoassets, however, is the matter of proving the legal owner. If a person cannot prove that crypto is or was theirs, then they cannot be protected by law, as the law cannot merely take their word for it. This is simple enough for most major exchanges, which according to Anti-Money-Laundering (AML) regulations require users to register their identity (In *Ion Science,* the court compelled an exchange to disclose information on the defendant); but for private crypto wallets the issue becomes more complicated. There is no registry of cryptoasset ownership, and enforcing one would be impossible, as all that is required to own cryptoassets is a string of characters that no-one else has used as a private key – and these can be generated with only a pencil and paper. According to *Possession, Relative Title, and Ownership in English Law*,[27] "the fact of possession gives rise to an alienable proprietary [i.e. ownership] interest that is not dependent on continued possession." Control of cryptoassets, and thus possession, is entirely determined by knowledge of the private key. Thus, cryptoassets appear to have similar properties to bearer bonds, which are securities owned by whomever possesses the physical certificate.[28] This raises issues when a private key is illegitimately stolen – how can a court be sure that someone is the legitimate legal owner of cryptoassets? How can a court rule if the ownership of cryptoassets is in dispute? Fortunately, there are methods the court could take to discover legitimate possession. A person could produce evidence that they were the ones in possession of the physical manifestation of the private key (for example, a wallet device, or a written record of the key) up until the ownership came into dispute. Another option is that a specialist could trace transactions from the public key and match

[24] *Ruscoe and another v Cryptopia Ltd (in liq)* [2020] NZHC 728, [2021] W.T.L.R. 965.
[25] *Ion Science Ltd v Persons Unknown* (HC, 28 January 2022).
[26] *Wang v Darby* [2021] EWHC 3054 (Comm), [2022] Bus. L.R. 121.
[27] Luke Rostill, *Possession, Relative Title, and Ownership in English Law* (OUP 2021).
[28] James Chen, 'Bearer Bond: Definition, How It Works, and Why They're Valuable' (*Investopedia*, 20 July 2021) <www.investopedia.com/terms/b/bearer_bond.asp> accessed 8 May 2023.

them with evidence provided by the person. Thus, while it would not be easy, it is possible to resolve ownership disputes in English law.

A further issue with the proprietary nature of cryptoassets is where they are considered to exist. This is important for questions of *lex situs* – the location the laws of which are applied to the asset. Most people who use cryptoassets hold them with international exchanges, so this is a pertinent issue. The precedent comes from *Ion Science*,[29] which held that the lex situs of a cryptoasset is the place where the owner of the cryptoasset is domiciled, which is where they have their permanent home.[30] The new proposals by His Majesty's Treasury (HMT) would encode this into regulation: "This would capture activities provided by UK firms to persons based in the UK or overseas (natural and legal), as well as those provided by overseas firms to UK persons (natural or legal)."[31] This makes sense from a consumer protection perspective – if an exchange is to provide services to people living in the UK, then it must obey British law in its relationships with them. This allows the UK the power to legislate and provide remedies to its citizens, and provides certainty to UK residents wishing to deal in cryptoassets – they can expect that, whichever corporation they buy their crypto from, that relationship will be governed by UK law and regulation. Per HMT,

> "Such an approach is in line with a number of other areas of the UK's regulatory perimeter and would likely be necessary since UK consumers can (and frequently do) easily access cryptoasset products and services which are provided by overseas companies. Furthermore, a definition which only captures firms operating in the UK would result in a situation where firms could move offshore to easily evade UK regulations but still serve UK customers."[32]

Thus to capture only firms operating in the UK would not only place consumers at risk, but also jeopardise the goal of making the UK a centre for the crypto industry.

[29] *Ion Science* (n 25).
[30] *Casdagli v Casdagli* [1918-19] All ER Rep 462, [1919] A.C. 145.
[31] HM Treasury, *Future financial services regulatory regime for cryptoassets* (2023).
[32] Ibid.

4. Risks to Consumers

As mentioned earlier, engaging with crypto presents a number of risks to the average consumer. For our purposes, these can be separated into two categories: risks exposed to by the use of crypto itself, and risks exposed to by the use of a crypto exchange.

The first risk – and perhaps the most obvious – is general financial risk arising from volatility. The price of crypto fluctuates a great amount, and investors often lose a great deal of money. Most recently, between the 30th of March and 19th of June 2022, the value of one Bitcoin dropped from $47,459.26 to $19,047.43, a loss of 60% in less than 3 months.[33] This is concerning given that 55% of people who own crypto do so as part of a long-term investment strategy.[34] The FCA warns that consumers should be prepared to lose *all* the money they are putting in.[35]

Secondly, fraud has surfaced in several high-profile 'crypto-scams' over the years, such as OneCoin and Bitconnect.[36] These take payments in established cryptoassets in order to escape moderation by the banking system. Because crypto transactions are anonymous and irreversible,[37] it is very difficult for law enforcement to reclaim the money. The leader of the OneCoin scandal, Ruja Ignatova, remains a fugitive and is currently on the FBI's 10 most wanted list.[38] According to the FBI, the scams themselves usually take the form of romance or confidence schemes;[39] the challenge of telling an honest investment to a fraudulent one is of the same nature as outside the crypto space, but is compounded by the difficulty in restituting the appropriated funds.

The risks arising from crypto exchanges deserve special attention because exchanges are the primary vector through which the public engages with cryptoassets. Three quarters

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[33] CoinMarketCap, 'Bitcoin Price' (*CoinMarketCap*) <https://coinmarketcap.com/currencies/bitcoin/> accessed 8 May 2023.
[34] Binance Research, '2021 Global Crypto User Index' (2021) <https://research.binance.com/static/pdf/Global_Crypto_Index_2021.pdf> accessed 8 May 2023.
[35] Financial Conduct Authority, 'Crypto: The basics' (*FCA*) <www.fca.org.uk/investsmart/crypto-basics> accessed 8 May 2023.
[36] Patty Ramsen, 'Worst Cryptocurrency Scams in History' *The Chain* (2018) <https://vocal.media/theChain/worst-cryptocurrency-scams-in-history> accessed 8 May 2023.
[37] Provided the wallet is self-custodied rather than held at an exchange.
[38] Federal Bureau of Investigation, 'Ten Most Wanted Fugitives' (*FBI*) <www.fbi.gov/wanted/topten> accessed 8 May 2023.
[39] Federal Bureau of Investigation, 'The FBI Warns of a Spike in Cryptocurrency Investment Schemes' (*Internet Crime Complaint Center*, 14 March 2023) <www.ic3.gov/Media/Y2023/PSA230314> accessed 8 May 2023.

of consumers purchase through an exchange,[40] and 60% of consumers hold their cryptoassets on an exchange.[41] There is a phrase common in the more experienced crypto community – "not your keys, not your coins."[42] This is because, as described above, control of an amount of cryptoassets is entirely determined by the keys associated with it. When one holds cryptoassets on an exchange, the keys to the crypto are held by the exchange, which grants access to the user through their account, and thus the user does not have ultimate control over their assets. This presents two key risks to a user of a crypto exchange: theft by hacking, and insolvency of the exchange.[43]

Hacking will always be a threat when one's private keys are stored digitally.[44] Exchanges, however, present a central point of attack. A hacker who is able to access or exploit the exchange's system can appropriate the crypto held therein. This has happened multiple times, most famously with the hacking of Mt. Gox: between the years of 2011-2014, 650,000 Bitcoin were stolen, primarily from the exchange's hot wallets (those used for the regular movement of assets) – a value today of approximately 14.3 billion USD. This led to the exchange halting withdrawals and shortly afterwards being liquidated by the Japanese government.[45] A more recent example is Ronin Network, from which 615 million USD worth of funds were stolen in March 2022, the second-largest crypto hack in history.[46] The reason these hacks are such a problem for the crypto industry is that, as mentioned before, crypto transactions are irreversible – there is no centralised banking or legal system to handle disputes. Control of cryptoassets is solely determined by possession of the private keys.

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[40] Financial Conduct Authority, 'Research Note: Cryptoasset consumer research 2021' (2021) <www.fca.org.uk/publications/research/research-note-cryptoasset-consumer-research-2021> accessed 8 May 2023.
[41] Binance Research (n 34).
[42] Kirsty Moreland, 'Not Your Keys, Not Your Coins. It's That Simple' (*Ledger Academy*, 4 September 2020) <www.ledger.com/academy/not-your-keys-not-your-coins-why-it-matters> accessed 8 May 2023.
[43] Additionally, if the exchange places restrictions on trading actions (e.g. for server maintenance) the user is forced to comply and may suffer losses as a result.
[44] Nathan Reiff, 'Can Crypto Be Hacked?' (*Investopedia*, 23 September 2022) <www.investopedia.com/articles/investing/032615/can-bitcoin-be-hacked.asp> accessed 8 May 2023.
[45] Alexandra Harney and Steve Stecklow, ' Twice burned - How Mt. Gox's bitcoin customers could lose again' *Reuters* (16 November 2017) <www.reuters.com/investigates/special-report/bitcoin-gox/> accessed 8 May 2023.
[46] Joe Tidy, 'A Ronin Network: What a $600m hack says about the state of crypto' *BBC News* (30 March 2022) <www.bbc.co.uk/news/technology-60933174> accessed 8 May 2023.

According to CipherTrace, only 20% of stolen crypto is ever recovered[47] (Compared to approximately 30% of general property[48]).

Insolvency is an issue in which the law plays a prominent role – there is a long established process for distributing a company's assets in such an event.[49] If an exchange goes out of business, what happens to the coins it was holding? When Lehman Brothers went bankrupt during the 2008 financial crisis, everyone who held accounts with them lost their money.[50] This is because depositing money at a bank is, legally, an unsecured loan – the bank owns the money, and the depositor owns the debt of the bank. If the bank becomes insolvent, the claims of unsecured creditors take the lowest priority (secured creditors first, then expenses, then preferential creditors, and then the general unsecured creditors[51]). In 2022, three major exchanges went bankrupt,[52] the most prominent of which was FTX, which will be discussed below. Following the Mt. Gox hack and subsequent insolvency, a significant number of stolen funds were recovered (totalling 200,000 Bitcoin). However, the Japanese courts ruled that the customers who had lost funds in the event were merely unsecured creditors, and so will only be paid out the remainder after higher priority charges are resolved.[53] If the UK is to promote consumer protection in the crypto markets, this is a preeminent issue to address.

Risks to consumers holding crypto at an exchange are compounded by the fact they do not have recourse to the Financial Services Compensation Scheme (FSCS).[54] The FSCS insures deposits and investments up to £85,000, compensating these if the firm they are held with

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[47] CipherTrace, 'Cryptocurrency Crime and Anti-Money Laundering Report, August 2021' (CipherTrace Cryptocurrency Intelligence, 2021) <https://ciphertrace.com/cryptocurrency-crime-and-anti-money-laundering-report-august-2021> accessed 8 May 2023.
[48] Statista, 'Recovery rate of stolen property in the United States in 2020, by type' (*Statista*, September 2021) <www.statista.com/statistics/252444/recovery-rate-of-stolen-property-in-the-us-by-type> accessed 8 May 2023.
[49] Elissa Jelowicki, 'How Assets are distributed to Creditors through Corporate Insolvency' (*Cross Channel Lawyers*, 25 February 2015) <www.crosschannellawyers.co.uk/how-assets-are-distributed-to-creditors-through-corporate-insolvency/> accessed 8 May 2023.
[50] *Re Lehman Brothers International (Europe) (In Administration) and Re the Insolvency Act 1986* [2012] UKSC6, [2012] 3 All E.R. 1.
[51] Jelowicki (n 49).
[52] Eric Rosenberg, 'What Happens When A Crypto Exchange Goes Bankrupt?' (*Investopedia*, 28 February 2023) <www.investopedia.com/crypto-bankruptcy-affecting-investors-6274367> accessed 8 May 2023.
[53] Michael Bacina, 'The long shadow of Mt. Gox' (*Piper Aldman*, 7 March 2019) <https://piperalderman.com.au/insight/the-long-shadow-of-mt-gox/> accessed 8 May 2023.
[54] Financial Conduct Authority, 'Investing in crypto' (*FCA*) <www.fca.org.uk/investsmart/investing-crypto> accessed 8 May 2023.

fails. However, the investment must be regulated by the FCA (or PRA) in order to qualify[55] – and, as mentioned above, exchange tokens are not. Even in the case that they become specified investments, HMT does not intend for exchange token holdings to be insured by the FSCS.[56]

Regulation of exchanges will be considered later, but first we will consider whether, in light of the above, and the potential for crypto to enable criminality, overall access to crypto should be restricted.

5. Restricting public access to Cryptoassets

In approaching the law of regulating activity, the central question is 'why should this be regulated' rather than 'why should it not be'? It is not the duty of a government to delineate the set of activities its populace are permitted to engage in but rather those which aren't. It would be ludicrously impractical at for the law to describe every permitted activity. The presumption is that the public can engage in an activity if they so choose, and it is on the onus of the law makers to have sufficient reason to interfere with that. This principle, that a person is entitled to act as they chose except as prohibited by law (commonly known as residual freedom), is an abiding tenet of common law.[57] This is an axiom of the societies of relative freedom we have enjoyed in the West.

So what constitutes sufficient reason? Lord Devlin argued that the law should enforce the common morality of society. As he wrote in *The Enforcement of Morals*, "For society is not something that is kept together physically; it is held together by the invisible bonds of common thought… The bondage is part of the price of society and mankind, which needs society, must pay its price."[58]

Due to the increasing adoption and lack of consensus, it is not clear that crypto use is against the common morality of society. One could point to certain coins' high use of energy (through their mining) as being contrary to society's environmental consciousness.[59] However, Ethereum recently enacted upgrades that reduced the network's energy

[55] Financial Services Compensation Scheme, 'Investment Compensation & Protection' (*FSCS*) <www.fscs.org.uk/what-we-cover/investments/> accessed 8 May 2023.
[56] HM Treasury, *Future financial services regulatory regime for cryptoassets* (2023).
[57] *Entick v Carrington* (1765) 2 Wils 275, 95 E.R. 807.
[58] Lord Devlin, *The Enforcement of Morals* (OUP 1965).
[59] Ephrat Livni, 'Can Crypto Go Green?' *New York Times* (11 October 2021) <www.nytimes.com/2021/10/10/business/dealbook/crypto-climate.html> accessed 8 May 2023.

consumption by 99.99%,[60] and in some cases crypto mining actually helps support renewable energy grids.[61]

Moreover, there are many reasons why someone might choose to honestly engage with cryptoassets. They can be more utile than conventional money, allowing one party to send to another anywhere in the world quickly[62] and with reduced fees.[63] This is the main reason why Nigeria has the highest rate of Bitcoin usage in the world at 32% - it is the preferred method of sending money back home.[64] Further, from a financial perspective, cryptoassets may provide a medium of exchange that is resistant to inflation. Built into their algorithms is a finite supply – for example, there will never be more than 21 million Bitcoins.[65] This could allow a popular cryptoasset to serve almost like the gold standard.[66] People may want to use crypto as an alternative to traditional financial institutions – 38% of crypto users cited distrust of the current financial system as a reason for owning crypto. In the 24 hours after the recent Silicon Valley Bank crash, Bitcoin rose 7.3%.[67] Further, a person may want to enjoy the privacy that crypto offers. Wanting to keep one's affairs private – even from the state – is a natural human inclination[68] even if one does not have 'anything to hide'.

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[60] —, 'Ethereum Blockchain Eliminates 99.99% of its Carbon Footprint Overnight After a Successful Merge According to New Report' *ConsenSys* (15 September 2022) <https://consensys.net/blog/press-release/ethereum-blockchain-eliminates-99-99-of-its-carbon-footprint-overnight-after-a-successful-merge-according-to-new-report> accessed 8 May 2023.

[61] Nic Carter, 'Bitcoin Mining Is Reshaping the Energy Sector and No One Is Talking About It' *CoinDesk* (11 October 2021) Opinion <www.coindesk.com/policy/2021/10/11/bitcoin-mining-is-reshaping-the-energy-sector-and-no-one-is-talking-about-it/> accessed 8 May 2023.

[62] Evander Smart, 'Blockchain Proven to be Faster Than Wire Transfers in Banking Payment Test' *Cointelegraph* (30 November 2016) <https://cointelegraph.com/news/blockchain-proven-to-be-faster-than-wire-transfers-in-banking-payment-test> accessed 8 May 2023.

[63] Joseph Young, 'Bitcoin is 300 Times Cheaper Than Wire Transfers, Banks Take 83% Profit' *NewsBTC* (2018) <www.newsbtc.com/news/bitcoin/bitcoin-is-300-times-cheaper-than-wire-transfers-banks-take-83-profit> accessed 8 May 2023.

[64] Ripples Nigeria, '32% Nigerians now use cryptocurrencies; Survey divides crypto experts' *Ripples Nigeria* (2 March 2021) <www.ripplesnigeria.com/32-nigerians-now-use-cryptocurrencies-survey-divides-crypto-experts/> accessed 8 May 2023.

[65] Bitcoins.net, 'Limits of Supply' (*Bitcoins.net*) <https://bitcoins.net/investing/limits-of-supply> accessed 8 May 2023.

[66] Crypto has recently tracked market falls as it is perceived as a high-risk investment, which means that it tends to fall as people become more risk-averse.

[67] Zahra Tayeb, 'Bitcoin rises and crypto market tops $1 trillion again after US regulators bail out SVB customers' *Insider* (13 March 2023) Markets Insider <https://markets.businessinsider.com/news/currencies/crypto-bitcoin-silicon-valley-bank-svb-customer-bailout-regulators-2023-3> accessed 8 May 2023.

[68] Jan Holvast, 'History of Privacy' (2009), The Future of Identity in the Information Society, IFIP Advances in Information and Communication Technology 298, 13-42.

Another view of when the law should step in comes from John Stuart Mill: "The only purpose for which power can be rightfully exercised over any member of a civilised community, against his will, is to prevent harm to others."[69] – in other words, to prevent harm to individuals or to society. Cryptoassets do pose a risk on a societal level in their enabling of criminal activity. We have already discussed why they make frauds and thefts easier to get away with, but there are two other very prominent areas to discuss.

The first is money laundering – the ability to transfer valuable assets instantaneously and anonymously has obvious uses[70] for malefactors wishing to hide money obtained in criminal activity or to be used in terrorist financing.[71] That said, although wallets are anonymous, transactions are public, meaning a dedicated member of the public can trace exactly to which addresses an amount of crypto has been transferred. However, the actual incidence of money laundering may not be as large as the threat first appears – according to one source, crypto transactions involving known illicit addresses made up only 0.15% of all transactions in 2021[72] (though there are likely more addresses related to illicit activity that are not known).

The second is tax evasion. An anonymous asset system poses problems for a government that wishes to tax income.[73] It is impossible to know how much crypto is owned by someone, and there is no way to prevent two parties privately exchanging crypto for goods or services. While transformations from the cryptoasset to fiat can easily be taxed as capital gains or income, transforming the cryptoasset into another cryptoasset, spending it directly on goods or services, or passively accruing value over time are much harder to monitor.

The privacy so central to the design of crypto is the primary issue for those concerned with stopping those who use crypto for criminal means. They may argue for prohibiting the ability for people to trade crypto anonymously, which would have to take the form of

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[69] John Stuart Mill, *On Liberty* (2, John W. Parker & Son 1860).
[70] Worldcoin, 'Understanding Money Laundering: How Common is it in Crypto?' (*Worldcoin*) <https://worldcoin.org/articles/cryptocurrency-money-laundering> accessed 8 May 2023.
[71] Institute of Financial Consultants, 'Anti-Money Laundering (AML)' (*IFA*) <www.ifa.org.uk/technical-resources/aml> accessed 8 May 2023.
[72] Chainalysis Team, 'Crypto Crime Trends for 2022: Illicit Transaction Activity Reaches All-Time High in Value, All-Time Low in Share of All Cryptocurrency Activity' (*Chainalysis*, 6 Jan 2022) <https://blog.chainalysis.com/reports/2022-crypto-crime-report-introduction/> accessed 8 May 2023.
[73] Greg Iacurci, 'Cryptocurrency poses a significant risk of tax evasion' *CNBC* (31 May 2021) Personal Finance <https://cnb.cx/3fSuctU> accessed 8 May 2023.

outlawing all private wallets (that is, wallets where the owner of the crypto controls their own private keys). In order to trade crypto, people who obey this law would have to use exchanges, who as per AML regulations require identification of all their users.[74]

This, however, would bring about its own issues. No longer able to look after their own crypto, law-abiding citizens would be forced to expose themselves to the risks present in using exchanges, described above. In response to a Bill recently introduced in the US that would require all wallet providers to obey AML regulation (thus removing the option of privacy), the Chamber of Digital Commerce stated, "Registering as financial institutions would impose significant compliance costs and hinder or preclude participation in the digital asset ecosystem. Covered entities may be forced to depart the U.S., resulting in a brain drain of talented developers and technical experts critical to continued digital asset ecosystem development."[75] Clearly, this move would thwart the objective of making the UK the "global centre for this activity" that the Treasury Committee described.

Furthermore, though requiring the use of exchanges would reduce AML risk, it would not stop all bad actors. The law deters criminality but does not stop it entirely, especially so in the case of crypto. Despite outlawing all crypto transactions in 2021, China still has 30 million crypto users.[76] Those who are determined to use crypto for dishonest purposes will do it whether or not the law tries to stop them. It is disproportionate to prevent law abiding citizens from using it in the attempt.

Returning to danger to the individual we come to the question of volatility – should the law seek measures to prevent people from investing in crypto so as to protect them from losing their money?

One is reminded of the old principle of English law, *caveat emptor*, or 'buyer beware'. The full phrase is '*caveat emptor, quia ignorare non debuit quod jus alienum emit'*, meaning 'let the buyer beware, for he ought not to be ignorant of the property which he is buying

[74] UK Government, *Factsheet: cryptoassets technical* (2023).
[75] Digital Chamber of Commerce, 'Statement on Digital Asset Anti-Money Laundering Act' (*Chamber of Digital Commerce*, 24 April 2023) <https://digitalchamber.org/statement-on-digital-asset-aml-act/> accessed 8 May 2023.
[76] Tsunami, 'China Has 30 Million Crypto Users Despite Ban' *AZCoin News* (15 April 2023) <https://azcoinnews.com/china-has-30-million-crypto-users-despite-ban.html> accessed 8 May 2023.

from another party.'[77] In other words, in cases where the principle applies the law accepts that the buyer bears the risks for whatever he is buying. Obviously this is not applied to every situation in the law, but we can draw on the reasoning. In the context of physical goods, *caveat emptor* means that the buyer has a responsibility to know the dangers and potential pitfalls of what he is purchasing. In the context of crypto, it would mean that the user has a responsibility to understand the risks at play.

Volatility is an expected result of an asset that is neither grounded in tangible property nor backed by a government. Thus, a buyer of crypto ought to accept the risks inherent in the technology, and a great deal of effort has been spent to apprise them of such. The law then accepts that they can take on that risk if they so choose. It is up to the buyer, not the law, to decide. This is the sort of choice that appears to me to be fully reasonable to permit in a lawful society.

6. Regulatory Categorisation of Cryptoassets

If the state does not seek to restrict access to crypto altogether, then it must figure out a regime to regulate it proportionately. The UK currently recognises three categories of cryptoassets as broadly outlined by the FCA – security tokens, e-money tokens, and exchange tokens. [78] The former two are regulated as "specified investments". The full meaning of this is beyond the scope of this essay, but in essence it means activities relating to such investments are regulated by the FCA.[79] Those performing them in the course of business must be registered with the FCA and are under numerous legal obligations. Specified investments include traditional financial devices such as shares.

Security tokens are those "that provide rights and obligations akin to specified investments".[80] These are shares, bonds, debentures, and other instruments representing an investment contract – an exchange of initial investment for the promise of future revenue, profit, ownership, voting rights, and the like. These tokens are regulated by the FCA in the same way as other specified investments. An example of this is the token SiaFund. SiaFund

[77] FindLaw, 'What does 'Caveat Emptor' Mean?' (*FindLaw*, 20 June 2016) <www.findlaw.com/consumer/consumer-transactions/what-does-caveat-emptor-mean-.html> accessed 8 May 2023.
[78] Financial Conduct Authority, *PS19/22: Guidance on Cryptoassets* (2019).
[79] Financial Conduct Authority, 'specified investment – FCA Handbook' (*FCA*) <www.handbook.fca.org.uk/handbook/glossary/G1117.html> accessed 8 May 2023.
[80] *Guidance on Cryptoassets* (n 78).

tokens were sold by the developer Nebulous to raise money, and in return holders of the tokens receive a portion of the company's transaction fees.[81] This token clearly is intended to mimic the function of traditional financial instruments, because it grants a right over the issuer in exchange for an investment, and thus it is sensible that they be regulated as such.

E-money is "electronically stored monetary value as represented by a claim on the electronic money issuer which is issued on receipt of funds for the purpose of making payment transactions, accepted by a person other than the electronic money issuer, [and] not excluded by regulation 3 of the EMRs."[82] Accordingly, E-money tokens are those the value of which is backed by the issuer, and are recognised as payment by someone other than the issuer – in a similar way to how fiat currencies are backed by a central bank. An example of this is the Poundtoken, which always tracks the value of the British pound as it is fully backed by its issuer, Blackfridge.[83] These are also regulated by the FCA. Again, this makes sense: the token is regulated as that which it imitates – in this case, money backed by a central issuer.

Exchange tokens are those that "are usually decentralised and primarily used as a means of exchange." They "provide limited or no rights for tokens holders, and there is usually not a single issuer to enforce rights against."[84] The value of these tokens lie within mutual recognition, rather than any right or backing. Unlike the above categories, they are currently not regulated by the FCA. These tokens form the majority of all crypto transactions – the two largest tokens in the world, Bitcoin and Ethereum, are both exchange tokens, and together account for over 60% of the cryptoasset market share.[85] Thus, at the moment the FCA does not regulate these exchange tokens. They are the cryptoassets intended to serve as a decentralised medium exchange, and how that fits into our existing system is in need of better understanding. This issue of how they should be legally categorised has been grappled with the world over. We can look at this question in general terms, and compare other countries' methods to the UK's.

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[81] Sia, 'What are Siafunds?' (*Sia Docs,* 21 December 2022) <https://docs.sia.tech/siafunds/what-are-siafunds> accessed 8 May 2023.
[82] *Guidance on Cryptoassets* (n 78).
[83] Oscar Hornstein, 'Poundtoken: Stablecoin tied to British pound launched by Blackfridge' *UKTN* (11 July 2022) <www.uktech.news/crypto/poundtoken-british-pound-stablecoin-blackfridge-20220711> accessed 8 May 2023.
[84] *Guidance on Cryptoassets* (n 78).
[85] TradingView, 'Cryptocurrency market' (*TradingView*) <www.tradingview.com/markets/cryptocurrencies/global-charts/> accessed 8 May 2023.

To start with, should they be classified as money? Two countries – The Central African Republic and El Salvador – have gone as far as making Bitcoin legal tender.[86] The colloquial term "cryptocurrency" implies that the assets are, indeed, currency. Traditionally, a currency is simply a medium of exchange, but in the modern era it has come to mean a token of promissory value.[87] A pound coin has value because the Bank of England promises to honour its value (and an e-money token because it is backed by its issuer), not because of mutual consensus – this is stark contrast to cryptoassets like Bitcoin or Ethereum, the value of which entirely stems from people's desire to use and invest in them. Thus, classifying exchange tokens as an actual currency makes little sense in today's world.

More attention has been directed towards classification as either a security or a commodity, typified by the ongoing rivalry between the USA's Securities and Exchange Commission (SEC),which regulates securities, and Commodity Futures Trading Commission (CFTC), which regulates commodities.[88] A common argument for classification as a security is that cryptocurrencies are commonly viewed as an investment opportunity by the public. Speaking in August 2021, the SEC Chair Gary Gensler said, "If somebody is raising money selling a token and the buyer is anticipating profits based on the efforts of that group to sponsor the seller, that fits into something that's a security."[89] This is comparable with the FCA's categorisation of security tokens, but does not fit for exchange tokens, and likewise the SEC have taken the view that Bitcoin and Ethereum are not securities.[90] These tokens are not an investment contract; they do not confer any rights or obligations on the holders.

Rather than being an investment contract, exchange tokens are a thing (granted, an incorporeal thing) that is bought and sold , or used to barter – much like a commodity. And indeed, this is the direction the US has taken, with Bitcoin now being classified as a

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[86] Yulia Volyntseva, 'Which Countries Have Adopted Bitcoin as Legal Tender' *Businesstech Weekly* (23 February 2023) <www.businesstechweekly.com/finance-and-accounting/cryptocurrency/which-countries-have-adopted-bitcoin-as-legal-tender/> accessed 8 May 2023.
[87] Ryan Thaxton, 'Fiat Money: What is a Fiat Currency?' (*FOREX,* 17 February 2023) <www.forex.com/en-ca/market-analysis/latest-research/what-are-fiat-currencies/> accessed 8 May 2023.
[88] Gina Chon, 'D.C. turf war opens crypto regulatory arbitrage' *Reuters* (18 August 2022) <www.reuters.com/breakingviews/dc-turf-war-opens-crypto-regulatory-arbitrage-2022-08-18/> accessed 8 May 2023.
[89] —, 'CNBC Exclusive: CNBC Transcript: SEC Chair Gary Gensler Speaks with CNBC's "Squawk Box" Today' *CNBC* (4 August 2021) <www.cnbc.com/2021/08/04/cnbc-exclusive-cnbc-transcript-sec-chair-gary-gensler-speaks-with-cnbcs-squawk-box-today.html> accessed 8 May 2023.
[90] Cassels, 'SEC Declares Bitcoin and Ether as Non-Securities' (*Cassels*, 26 June 2018) <https://cassels.com/insights/sec-declares-bitcoin-and-ether-as-non-securities/> accessed 8 May 2023.

commodity, and Ethereum likely to be taken in the same direction.[91] A good analogy for exchange tokens is gold – both have a limited supply, both hold value primarily due to mutual consensus rather than practical use, and in the past, gold was even used as a medium of exchange.[92] In fact, this is also the approach the UK has taken; commodities, like exchange tokens, are not regulated by the FCA.[93]

In their guidance, the FCA say that they do not regulate crypto much like they do not regulate people trading in fine art or wine.[94] Outside of tax, illegal substances, and unfair practices (such as misleading or aggressive actions[95]), the law does not restrict 'bartering', the practice of individuals trading things for other things. Within the vision of crypto as a 'commodity', then, it makes sense that exchange tokens remain unregulated. They would, in the eyes of the law, take the role of a thing to be bought, sold, and traded, which would match well their intended use. International legal experts from outside the UK have also acknowledged that the UK treats exchange tokens like commodities.[96]

However, in their 2023 consultation paper, HMT has set out plans to bring all cryptoassets under the umbrella of specified investments.[97] This is contrary to the FCA's stated view in their cryptoasset guidance – as discussed previously, a "commodity" does not come under this definitional bracket. Those that do are either investment contracts (e.g. shares, bonds) or money-related instruments (e-money, deposits). In the words of the FCA:

> "We are aware that exchange tokens can be acquired and held for the purpose of speculation rather than exchange, as token-holders may anticipate that the value of these tokens will increase in the future on cryptoasset markets. However, we do not view this as being sufficient for exchange tokens to constitute specified investments.

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[91] Luke Huigsloot, 'CFTC declares Ether as a commodity again in court filing' *Cointelegraph* (14 Dec 2022) <https://cointelegraph.com/news/cftc-declares-ether-as-a-commodity-again-in-court-filing> accessed 8 May 2023.
[92] The Editors of Encyclopaedia Britannica, 'Gold standard', Encyclopaedia Britannica <www.britannica.com/topic/gold-standard> accessed 8 May 2023.
[93] Futures and other derivatives of commodities are, however, considered to be specified investments, and likewise futures and other derivatives of exchange tokens.
[94] *Guidance on Cryptoassets* (n 78) [45].
[95] The Consumer Protection from Unfair Trading Regulations 2008.
[96] Daniel Holman and Barbara Stettner , 'Anti-Money Laundering Regulation of Cryptocurrency: U.S. and Global Approaches' (Allen & Overly LLP, 2018) <www.allenovery.com/global/-/media/sharepoint/publications/publications/en-gb/documents/aml18_allenovery.pdf> accessed 8 May 2023.
[97] HM Treasury, *Future financial services regulatory regime for cryptoassets* (2023).

> The analogy would be an individual holding different fiat currency or a commodity, both of which are unregulated, in the hope of a gain."[98]

By this reasoning exchange tokens do not belong under the definition of a specified investment.

Furthermore, from a practical perspective, a lack of regulation on the trading of cryptoassets allows for them to fully serve their intended purpose as a decentralised medium of exchange. It is well established that regulation increases costs on firms, and that those costs pass on to consumers. It is currently not at all feasible for a member of the public to easily exchange shares between themselves[99] (for example, to use as payment). Cryptoassets, in particular exchange tokens, are designed to be sent quickly, frequently, and easily. Stifling this would hinder the nascent crypto industry. HMT concurs with this approach, stating that it "and the financial services regulators should avoid applying disproportionate or overly burdensome regulation to entities."

As a final point, regulating exchange tokens as investments may be premature. Though their current use and perception does not strongly match up with their intent, that may change in the future. According to one source, 184 online companies accept Bitcoin as payment in the UK.[100] In addition, it can be used indirectly with larger retailers, such as Tesco or Primark, through the use of third party gift certificates or licence fee stores.[101] Furthermore, adoption of crypto as a medium of exchange may increase as their volatility decreases (which it is doing[102]), and thus their price becomes more reliable.

HMT's proposals do allow some leeway for the FCA to modulate the regime specific for crypto:

> "The FCA's general rule making powers would be available, allowing the FCA to design regulatory regimes for newly added activities. In practice this means that the FCA will be given powers to write tailored rules, as opposed to the existing rules automatically

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[98] *Guidance on Cryptoassets* (n 78) [42].
[99] Deposits are faster due to cooperation between the banks.
[100] Where To Spend Bitcoins UK, 'Online retailers in the UK that accept bitcoins' (*wheretospendbitcoins*) <www.wheretospendbitcoins.co.uk/region/online.html> accessed 8 May 2023.
[101] Brean Horne, Companies that Accept Bitcoin in the UK (*nerdwallet*, 24 June 2022) <www.nerdwallet.com/uk/business/companies-that-accept-bitcoin/> accessed 8 May 2023.
[102] Bitcoinist, 'Looking at Bitcoin Volatility Over the Years' (*Bitcoinist,* 2020) <https://bitcoinist.com/looking-at-bitcoin-volatility-over-the-years/> accessed 8 May 2023.

applying to cryptoassets. The FCA will need to consider what is appropriate and consult accordingly."[103]

One hopes, therefore, that should these proposals move forward the FCA will indeed take this opportunity to write tailored rules. A more permissive regime with regards to the actual movement of crypto (and access to the movement) is more important for the crypto industry than it would be for traditional finance. We will now discuss the unique challenges that apply to the firms themselves.

7. Regulation of Crypto Exchanges

Here is a sample of the UK-applicable user agreement of Coinbase, the largest crypto exchange operating within the UK:

"5.19 ***Digital Currency Title.*** *All Digital Currencies held in your Digital Currency Wallet are assets held by the Coinbase Group for your benefit on a custodial basis. Among other things, this means:*

> *(A) Title to Digital Currency shall at all times remain with you and shall not transfer to any company in the Coinbase Group…*
>
> *(B) None of the Digital Currencies in your Digital Currency Wallet are the property of, or shall or may be loaned to, Coinbase; Coinbase does not represent or treat assets in a user's Digital Currency Wallets as belonging to Coinbase... Coinbase will not sell, transfer, loan, hypothecate, or otherwise alienate Digital Currency in your Digital Currency Wallet unless instructed by you or compelled by a court of competent jurisdiction to do so."*[104]

Coinbase describes itself as undertaking the role of a 'custodian'. This is a commonly understood term in the legal sector, and refers to an "institution that holds customers' securities for safekeeping".[105] Coinbase then goes on to explain the specifics – primarily, the user retains legal ownership of the assets at all times (clause A), with Coinbase 'safeguarding' them. This is different from a trust, in which the beneficiary possesses

[103] *Future financial services regulatory regime for cryptoassets* (n 97) [2.7].
[104] Coinbase, 'User Agreement' (*Coinbase*) <www.coinbase.com/legal/user_agreement/payments_europ accessed 8 May 2023.
[105] Adam Barone, 'Custodian: What It Means in Banking and Finance' (*Investopedia*, 18 September 2022) <www.investopedia.com/terms/c/custodian.asp> accessed 8 May 2023.

equitable interest while the trustee owns legal title. It is also, importantly, different from a bank deposit, in which the assets (in this case, money) is *loaned* to the bank.[106] A bank is not a custodian, but receives the money into its ownership as a debtor; the customer is not an owner of money but an unsecured creditor without proprietary rights over their loans.[107] This means that, in the event that the bank becomes insolvent, their claim will only be contractual rather than proprietary, placing them at the bottom of the priority list in an insolvency, although the FSCS will insure deposits up to £85,000.

With the custodial system, however, the user retains their proprietary interest in the assets, so the assets do not appear on the firm's balance sheets. [108] They would therefore not be swept up in the event of an insolvency, meaning they are returned to the owner and never become part of the property that is distributed according to the hierarchy of creditors.[109]

Furthermore, in clause B Coinbase states that they "will not sell, transfer, loan, hypothecate, or otherwise alienate Digital Currency… unless instructed by you or compelled by a court of competent jurisdiction…" The consequence of all this is that the Coinbase will retain in its custody all the assets it is safeguarding, so even in the event of a 'bank run' Coinbase will be able to match all withdrawals. The assets never appear on Coinbase's balance sheets, so even in the event of insolvency they will be restorable in full to the customers.[110]

This system has obvious desirable qualities to consumers, as they can trust their assets would be safe in the event of an insolvency or bank run (presuming all rules are adhered to, which will be discussed shortly). However, it does come at a cost – rather than being able to leverage the assets in its custody, Coinbase makes most of its money from transaction fees.[111] This means there is a higher cost to trading than on other exchanges.[112]

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[106] Jeanne Gobat, 'What is a Bank?' (*International Monetary Fund,* March 2012) <www.imf.org/en/Publications/fandd/issues/Series/Back-to-Basics/Banks> accessed 8 May 2023.
[107] *Re Lehman Brothers International (Europe) (In Administration) and Re the Insolvency Act 1986* [2012] UKSC6, [2012] 3 All E.R. 1.
[108] Financial Conduct Authority, *Client Asset Sourcebook* (2023).
[109] Ibid [5.5].
[110] Ibid.
[111] Soocial, 'How Does Coinbase Make Money?' (*Soocial*) <www.soocial.com/how-does-coinbase-make-money/> accessed 8 May 2023.
[112] Chandan Lodha, '2022 Crypto-Exchange Fee Comparison' (*CoinTracker*, 22 July 2022) <www.cointracker.io/blog/2019-crypto-exchange-fee-comparison> accessed 8 May 2023.

An exchange which places fewer restrictions on itself would also be able to offer lower fees to its customers. If the firm did alienate a user from their assets (in order to make money by market making, for example), then the user would no longer have a proprietary interest in the assets as the legal title would not be theirs. Rather, they would have a contractual interest, making them unsecured creditors, and would be subject to the counterparty risk of the exchange's insolvency.[113] Should the public be able to choose this option, at the cost of greater risk to their assets? As Charles Randell, Chair of the FCA, said, "we need to strike the right balance between fostering innovation, providing an appropriate level of protection and allowing individuals freedom to take decisions for which they are responsible."[114]

An obvious solution here is the concept of informed consent. That is, the idea that a person should be able to accept risks as long as they are aware of what such risks are. This is something the FCA already utilises when regulating information provided to the customer.[115] The key risks here are (1) that the user does not retain title of their assets, and so would lose them in an insolvency of the exchange; and (2) that the exchange retains the ability to loan or otherwise alienate the assets from the consumer, meaning that the assets may not be there for the consumer to withdraw should they so desire. Is this something the law should see as acceptable?

In fact, this is something the law already allows in a related industry, as it is similar to stocklending as performed by a stock broker.[116] As an industry standard, brokers reserve the right to lend shares to the market (typically short sellers) for their own purposes.[117] The customers relinquish their proprietary interest in return for the broker's contractual promise, and thus becomes subject to the risk of insolvency (as their claim is contractual

[113] Hossein Nabilou, 'The Law and Macroeconomics of Custody and Asset Segregation Rules: Defining the Perimeters of Crypto-banking' (*Oxford Business Law* Blog, 13 April 2022) <https://blogs.law.ox.ac.uk/business-law-blog/blog/2022/04/law-and-macroeconomics-custody-and-asset-segregation-rules-defining> accessed 8 May 2023.
[114] Charles Randell, 'The risks of token regulation' (Speech to the Cambridge International Symposium on Economic Crime, Cambridge, 6 September 2021) <www.fca.org.uk/news/speeches/risks-token-regulation> accessed 8 May 2023.
[115] Financial Conduct Authority, *Conduct of Business Sourcebook* [4.1A].
[116] Thomson Reuters, 'Stock lending' (*Thomson Reuters Practical* Law) <https://uk.practicallaw.thomsonreuters.com/4-107-7322> accessed 8 May 2023.
[117] Blackrock, 'What is Securities Lending?' (*iShares*) <www.ishares.com/uk/professional/en/education/securities-lending> accessed 8 May 2023.

rather than proprietary, they are creditors and their assets become swept up in the insolvency).[118]

If a crypto firm chooses to use this 'alienation' model, it would be material information for the consumer. A person may reasonably expect that the cryptoassets held at the account were 'theirs', and would want to be notified if this were not true. In the spirit of informed consent, legislation could draw inspiration from the Unfair Contract Terms Act (UCTA).[119] Legislation could require firms who do this to draw particular attention to this, in the same way that the UCTA requires particular attention to be drawn to certain onerous contract terms. This would prevent such an important term from being hidden amidst the terms and conditions, and thus allow consumers to make an informed choice on the risks they are willing to undertake.

Furthermore, just because an exchange's user agreement contains language to the above extent, does not mean that it will abide by such agreements. FTX contained much of the same language as Coinbase does. Though the FTX website is no longer available, fortune.com reports that the US terms of agreement stated that: "Title to cryptocurrency represented in your FTX.US Account shall at all times remain with you and shall not transfer to FTX.US"; and that "You control the Digital Assets held in your Account. None of the Digital Assets in your Account are the property of, or shall or may be loaned to, FTX Trading".[120]

However, that is not what FTX did in practice. 10 billion dollars' worth of user cryptoassets were reportedly transferred to Alameda Research, another company owned by FTX's primary owner, Sam Bankman-Fried.[121] This was contrary to FTX's terms of use, and because customers retained ownership of the funds, also alleged criminal theft.

While this is a US case, the principle here is universal. Just because an act is illegal, if a party thinks they can get away with it, then they may try it anyway. In order to have more

[118] Financial Conduct Authority, *Collective Investment Schemes Sourcebook* [5.4].
[119] Unfair Contract Terms Act 1977.
[120] Megan Leonhardt, 'FTX users have a leg up on other crypto bankruptcy victims—on paper' *Fortune* (17 November 2022) Crypto <https://fortune.com/crypto/2022/11/17/ftx-users-may-be-in-a-better-position-to-reclaim-their-assets-than-in-previous-crypto-bankruptcies/> accessed 8 May 2023.
[121] Vicky Ge Huang and others, 'FTX Tapped Into Customer Accounts to Fund Risky Bets, Setting Up Its Downfall' *The Wall Street Journal* (11 November 2022) <www.wsj.com/articles/ftx-tapped-into-customer-accounts-to-fund-risky-bets-setting-up-its-downfall-11668093732> accessed 8 May 2023.

preventative structures in place, the law must prescribe rather than just prohibit.[122] By enforcing proper procedures and conduct, the law makes it much more difficult for would-be malefactors to act dishonestly than merely making such actions illegal. But if too many procedures are prescribed, then honest corporations may be unduly hindered by additional costs and we risk losing the prosperous free market we are aiming for.

A useful tool in the regulator's pocket is the audit. The FCA currently requires several types of financial firms to submit to internal or external audits.[123] This is to ensure that such firms follow due procedure. A firm that promises not to alienate a client's assets could be required to submit to an audit to ensure this is so, in order to prevent a repeat of the FTX debacle.

This places a burden on firms that promise that their customers will retain legal title of the cryptoassets. Those that choose the 'alienation' model should not be granted a lighter regulatory burden for exposing their users to more risk. Rather than conduct requirements, which governs the specific practices of a firm, prudential requirements, which require firms to act in a prudent way,[124] may be more appropriate. The Bank of England states the role of prudential requirements is "to ensure that firms act safely and reduce the chance of getting into financial difficulty"[125] because "when a bank fails… people and businesses can lose money they have placed in a bank."[126] This parallels crypto firms following the 'alienation' model. The FCA, despite the name, is also equipped to apply prudential regulation. Comprehensive prudential recommendations are a question of good business management. Nausicaa Delfas, director of Specialist Supervision of the FCA, describes a regular review to "determine the appropriate level of capital and liquidity that the firm should hold… a thorough assessment of a firm's risk management exposures, management capability and controls."[127]

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[122] Financial Conduct Authority, 'About the FCA' (*FCA,* 19 April 2016) <www.fca.org.uk/about/what-we-do/the-fca> accessed 8 May 2023.
[123] Financial Conduct Authority, *Supervision Manual* [3.1].
[124] Bank of England, 'Why do we regulate banks?' (*Bank of England,* 17 June 2019) <www.bankofengland.co.uk/explainers/why-do-we-regulate-banks> accessed 8 May 2023.
[125] Bank of England, 'What is the Prudential Regulation Authority (PRA)?' (*Bank of England*, 15 October 2020) <www.bankofengland.co.uk/explainers/what-is-the-prudential-regulation-authority-pra> accessed 8 May 2023.
[126] 'Why do we regulate banks?' (n 124).
[127] Nausicaa Delfas, 'Overview of the FCA prudential approach' (Speech at the first FCA Prudential Supervision Forum, 21 May 2015) <www.fca.org.uk/news/speeches/overview-fca-prudential-approach> accessed 8 May 2023.

In section 4, I highlighted the risk of hacking when dealing with exchanges. Hacking has largely come to target "decentralised finance" (DeFi) protocols – in other words, attacking transaction code directly. 80% of crypto stolen in 2022 was from hacking DeFi protocols.[128] Exchanges are still a focus in this – in the largest hack in 2022, £500 million was stolen from the RONIN exchange via a vulnerability in one such DeFi protocol.[129] DeFi is a newly emerging and rapidly evolving space. Developers for these protocols are decentralised, quite possibly being scattered around the globe, so there is no practical way English law can control their development or their use.

Ronhui Gu, CEO of Certik, a blockchain security firm, said that every line of code should be audited before going live.[130] The law could demand that exchanges, who have been common hacking targets in the past, take measures to protect their accounts, which they ought to be doing anyway. One possibility is to imply a duty to take reasonable lengths to safeguard consumer assets from hacking: A crypto firm submitting to an audit would have to show they have taken reasonable steps to ensure their cybersecurity. And, because the hack of a firm using the lending model would also cost user their investment, this requirement would preferably apply to them as well.

As a final point on hacking, exchanges that do not alienate users from their assets would be safer for the consumers in the event of a hack. As mentioned previously, the customers who lost funds in the Mt Gox hack and insolvency are considered to be unsecured creditors by Japanese law, so are likely to see very little of their original investment back.

It is beneficial, therefore, for customers to have the option of using the safer, custodial model. To keep this model economical (both for firms, and for users), the regulatory burden should be kept light (recall that more regulation increases costs). The FCA currently applies extensive custodial requirements to regulated firms who hold applicable investments on behalf of users – the Client Asset Sourcebook regime (CASS).[131] Under the

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[128] Tim Fries, 'A Record $3.7B Was Stolen in Crypto Hacks in 2022, DeFi Most Targeted' *The Tokenist* (30 December 2022) <https://tokenist.com/a-record-3-7b-was-stolen-in-crypto-hacks-in-2022-defi-most-targeted/> 8 May 2023.

[129] Andrew Thurman, 'Axie Infinity's Ronin Network Suffers $625M Exploit' *CoinDesk* (29 March 2022) <www.coindesk.com/tech/2022/03/29/axie-infinitys-ronin-network-suffers-625m-exploit/> accessed 8 May 2023.

[130] Lindsey Choo, 'Crypto is crumbling, and DeFi hacks are getting worse' *protocol* (16 May 2022) <www.protocol.com/fintech/defi-hacks-web3> accessed 8 May 2023.

[131] Financial Conduct Authority, *Client Asset Sourcebook.*

government's new proposals, CASS would be the foundation for the regime for all cryptoassets.[132] In order to keep costs down, a more lightweight regime may be more appropriate. The above proposals (clear contract terms, implied duties, and audits) could be a good starting point.

8. Conclusion

The Cryptoasset industry is booming and, while superficially similar to traditional financial systems, it would benefit from a tailored legal or regulatory regime that strives to remain unburdensome in order to keep costs down. This would allow the potential for crypto to fulfil its purpose as a medium of exchange, and foster a free and accessible market. I submit the following measures to further consumer protection in light of this:

1. Exchange tokens to remain unregulated by the FCA, or at least to receive a tailored regime that is lower in burden than that for traditional financial instruments.
2. Firms that act on a custodial basis for customers to submit to an auditing system to ensure customer assets are held as promised.
3. Firms that alienate assets from their users to be required to draw particular attention to the terms in their user agreement that facilitate this.
    a. In addition, to be subject to prudential review to ensure good risk management.
4. Firms of both types to be subject to a duty to take reasonable steps to prevent theft by hacking.

These measures constitute a proportionate approach to protecting consumers and society while still fostering a free and accessible market. While the future is and always will be uncertain, they may a solid foundation on which to proceed.

[132] HM Treasury, *Future financial services regulatory regime for cryptoassets* (2023).

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